Question 1 (10 Points)

Create a Pull Request and Create a Plan

Notes: To do it off the latest main branch use the following commands. You can also change main to another branch if you want to reuse previous work.

```
git checkout main
git pull
git checkout -b <branch>
<edit some file>
git add <file>
git commit -m “Homework assignment X”
git push origin <branch>
<Go to Github and be sure to create the PR>
```

Create a comment for your PR with a plan of action. Your plan of action should encompass how you plan to finish each of the other questions in this homework assignment. Do not edit this comment and it should be the first comment in the PR history.

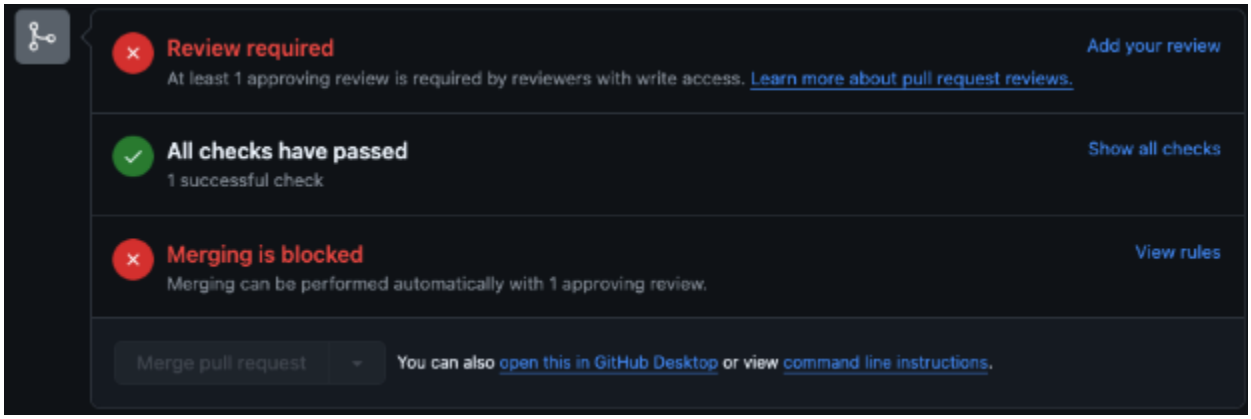
Abraham Lincoln is often quoted as saying, “If I had eight hours to chop down a tree, I'd spend the first six hours sharpening my axe”. You will be graded based on the preparation and research you put into it. Your plan of action should be a short essay.

In your plan of action explain various approaches that you could take. Talk about their potential pros and cons. What alternatives did you consider to these decisions?

If it helps to think in these terms, this is essentially a mini-ADR (Architectural Decision Record).
Resource:
https://github.com/joelparkerhenderson/architecture-decision-record
Template:
https://github.com/joelparkerhenderson/architecture-decision-record/tree/main/locales/en/templates/decision-record-template-by-michael-nygard

After creating the PR, since you haven’t made any major changes yet, you should see that the testing pipeline ran and passed at the bottom of the PR. After completing the assignment, you’ll

need to ensure this is still green to get full points.



Question 2 (5 Points)

Train a model

Some resources you might want to check out while working on this assignment:

Huggingface Finetuning
https://huggingface.co/docs/transformers/training
With Sagemaker
https://huggingface.co/docs/sagemaker/train

Unsloth Notebooks:
Conversation:
https://colab.research.google.com/github/unslothai/notebooks/blob/main/nb/Llama3.2_(1B_and_3B)-Conversational.ipynb
Alpaca:
https://colab.research.google.com/github/unslothai/notebooks/blob/main/nb/Llama3.1_(8B)-Alpaca.ipynb

Create a trainer.py file in your src directory. For training, I recommend you use your data loader and evaluator from previous homework assignments. Note: if your data loader and evaluator are using the same underlying dataset, you'll need to resolve that conflict to prevent data leakage issues.

For most students this will be your first time training an LLM, I highly recommend using Llama 3.2 1B as your base.

You will need a GPU to use unsloth, so using your dev env setup from HW 4 is recommended. You will also have to account for this in your tests as your CI/CD will likely fail since it won't have

a GPU.

Run your evaluator before and after training and post the results in a comment in your PR. How did your model do? Did it improve at your task? What other datasets are available to train your model on to help it improve at your chosen task? How long did it take to train? How much VRAM did it take to train? How much did the loss improve? What does that mean?

Question 3 (3 points)

Write tests and format your code

In the tests directory create file “test_trainer.py”.

Using pytest write as many tests as needed to show your Trainer is working as intended.

When you are done, format your code with ruff.

Add and commit your code and push it to your PR.

Ensure the CI test pipeline passes!

Question 4 (2 points)

Explain your work

Review your PR. Add any helpful notes inline to your code explaining your work. In a comment be sure to explain how things went. Did things go as you initially planned? Why or why not?

Explain why you wrote the tests you did.

If you had more time, what would you do to improve this PR?